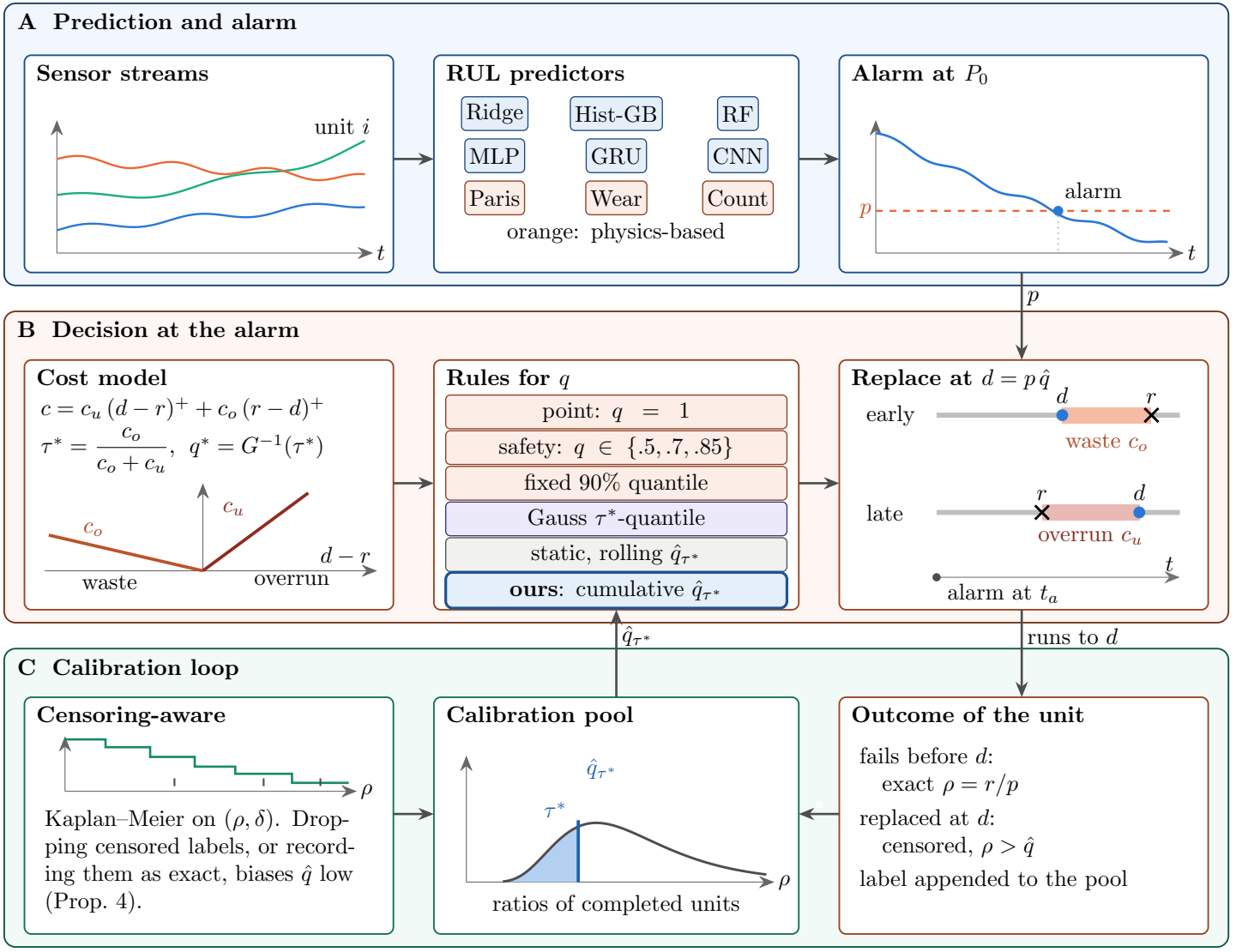

A Prediction and alarm
Sensor streams
unit i
t
RUL predictors
Ridge
Hist-GB
RF
MLP
GRU
CNN
Paris
Wear
Count
orange: physics-based
Alarm at P_0
p
alarm
t
p
B Decision at the alarm
Cost model
$c = c_u\,(d-r)^+ + c_o\,(r-d)^+$
$\tau^* = \dfrac{c_o}{c_o + c_u},\ \ q^* = G^{-1}(\tau^*)$
c_o
c_u
$d - r$
waste
overrun
Rules for q
point: $q = 1$
safety: $q \in \{.5, .7, .85\}$
fixed 90% quantile
Gauss τ^*-quantile
static, rolling $\hat{q}_{\tau^*}$
ours: cumulative $\hat{q}_{\tau^*}$
Replace at $d = p\,\hat{q}$
early
d
r
waste c_o
late
r
d
overrun c_u
t
alarm at t_a
$\hat{q}_{\tau^*}$
runs to d
C Calibration loop
Censoring-aware
ρ
Kaplan–Meier on (ρ, δ). Dropping censored labels, or recording them as exact, biases $\hat{q}$ low (Prop. 4).
Calibration pool
$\hat{q}_{\tau^*}$
τ^*
ρ
ratios of completed units
Outcome of the unit
fails before d:
exact $\rho = r/p$
replaced at d:
censored, $\rho > \hat{q}$
label appended to the pool